Mahitha Karnati

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EDUCATION

University of Michigan Ann Arbor, MI
Bachelor of Science in Engineering, Data Science May 2026

Relevant Courses: Computational Modeling Tools and Techniques, Data Structures and Algorithms, Applied Regression Analysis, Discrete Mathematics, Database Management, Introduction to Machine Learning, Web Design and Development, Building Data-Driven Applications, Introduction to Artificial Intelligence

SKILLS

Programming Languages: C++, Python, SQL, Java, R, HTML, JavaScript, C, C#
Software Tools: RStudio, Jupyter Notebook, Microsoft Excel, Oracle, Vim, Git, REST API, JSON, Xcode, VS Code, Django, MongoDB, GCP, GitLab, Tableau, Power BI

PROFESSIONAL EXPERIENCE

Forvia HELLA Northville, MI
Data Design and Development Intern June 2025 — August 2025
- Automated a manual scheduling process into a scalable system, cutting planning time and improving resource allocation across 5 teams
- Engineered a Python ETL pipeline to extract and transform project and workload data from company databases, automating back scheduling to compute phase start dates from fixed production deadlines
- Created a dashboard with interactive Gantt charts showing project timelines, workload distribution, and schedules
- Integrated AI driven agents to automatically refresh data and let users adjust planning scenarios, improving interactivity and reducing manual adjustments and planning by over 30+ hours per month

Youphoria Remote
Technology Specialist May 2025 — Present
- Enhanced user interface for an application that is focused on promoting mental health and wellness
- Developed and integrated a Firebase backed push notification system for likes and comments to improve interactivity which increased recent user engagement metrics by 30%
- Built a weekly leaderboard with pinned user view and integrated nudge functionality, handling API rate limit errors

NITS Solutions Novi, MI
Data Analyst Intern June 2024 — August 2024
- Queried and processed 500K+ records from multiple SQL databases to build structured datasets for modeling electric vehicle trends for Audi and Volkswagen
- Optimized SQL queries, reducing runtime by 20% and streamlining data preparation for analytical workflows
- Developed and trained predictive models in Python to forecast electric vehicle sales patterns and evaluated performance using statistical and machine learning metrics
- Delivered interactive data visualizations to communicate key insights and support business decisions to management

PROJECTS

Netflix Recommendation System
Languages and Tools Utilized: Python,Jupyter Notebook, API, Git, Scikit-Learn, NumPy
- Built a Python application for personalized Netflix recommendations based on user's viewing history
- Conducted an EDA and implemented collaborative filtering and content based recommendation algorithms to improve personalization
- Evaluated performance with RMSE and precision recall scores and created reports detailing customer insights and effectiveness

Stock Market Simulation
Languages and Tools Utilized: C++,Git, VS Code,STL
- Built a simulation of a stock exchange platform, handling real-time buy and sell order processing and matching
- Validated structured inputs (timestamp, trader/stock IDs, side, price, quantity) to ensure data integrity
- Implemented and compared priority queue structures for performance evaluation

Predicting ICU Survival
Languages and Tools Utilized: Python, Scikit-Learn, Git, Numpy, Matplotlib
- Engineered patient level features from 48 hour ICU time series and applied imputation and normalization to create modeling inputs
- Trained and tuned L1 and L2 logistic regression models with stratified 5-fold CV, optimizing hyperparameters for AUROC and F1
- Implemented weighted classes and threshold adjustments to address imbalances and evaluated models with ROC curves, precision-recall curves, and confusion matrices
- Compared logistic regression and kernel ridge models using RBF kernels to evaluate nonlinear gains while maintaining interpretability

CAMPUS INVOLVEMENT

Michigan Data Science Team Ann Arbor, MI
Project Team Member January 2024 — Present
Languages and Tools Utilized: Python, PyTorch, Scikit-learn, Git, Jupyter Notebook
- Worked on a program to create an automated spymaster for a word association game called Codenames
- Created a word2vec model that represents the words as vectors, shows similarity scores, and applied K-means clustering and Q-learning to improve the model and produce better results